

Compiler Design Implementation

Task 1: Lexical Analyzer Programming

Task 2: Syntax Parser Programming

Notices:

(1)The programming language is not limited (better C or Java).

(2)You can finish the labs in any method in the suggested optional methods.

(3)The complexity of lexical and syntactic definitions is decided by yourself.

Requirements on Task 1

(1)Input

Stream of characters

REs(The number of REs is decided by yourself)

(2)Output

Sequence of tokens

(3)Classes of words are defined by yourself

(4>Error handling may be included

*Optional Implementing Methods on Task 1

- (1) Simulating programs in Section 2.6.5
- (2) Programming based on FA
 - (a) Define some REs by yourself
 - (b) Convert REs into NFAs
 - (c) Merge these NFAs into a single NFA
 - (d) Convert the NFA into a DFA^o with minimum states
 - (e) Programming based on the DFA^o
- (3) Generating programs based on your own Lex
 - (a) Design your own Lex
 - (i) Define .l file
 - (ii) Programming your own Lex
 - (b) Generating programs based on your own Lex

Requirements on Task 2

(1)Input

Stream of characters

CFG(Combination of CFGs of some classes of sentences)

(2)Output

Sequence of derivations if top-down syntax analyzing methods are used.

Sequence of reductions if bottom-up syntax analyzing methods are used.

(3)Classes of sentences are defined by yourself

(4>Error handling may be included

***Optional Implementing Methods on Task 2**

(1) Recursive descent parsing

A non-terminal is related to a function/ sub-routine

(2) LL(1)

(a) Construct LL(1) parsing table based on the CFG

(b) Design the program using LL(1) parsing table

(3) LR(1)

(a) Construct LR(1) parsing table based on the CFG

(b) Design the program using LR(1) parsing table

Lab Report & Source Code Requirements

(1) Lab Report's Contents Must Contain:

- (a) Motivation/Aim
- (b) Content description
- (c) Ideas/Methods
- (d) Assumptions
- (e) Related FA descriptions
- (f) Description of important Data Structures
- (g) Description of core Algorithms
- (h) Use cases on running
- (i) Problems occurred and related solutions
- (j) Your feelings and comments

(2) Source Code

- (a) A CD is needed **by a class in the end of this terms**
- (b) Each person is related to a directory in the CD (StudentID_Name)
- (c) In the directory of a person, the follow files are needed:

Input file, Output file, Source program files, Report file, Other related files